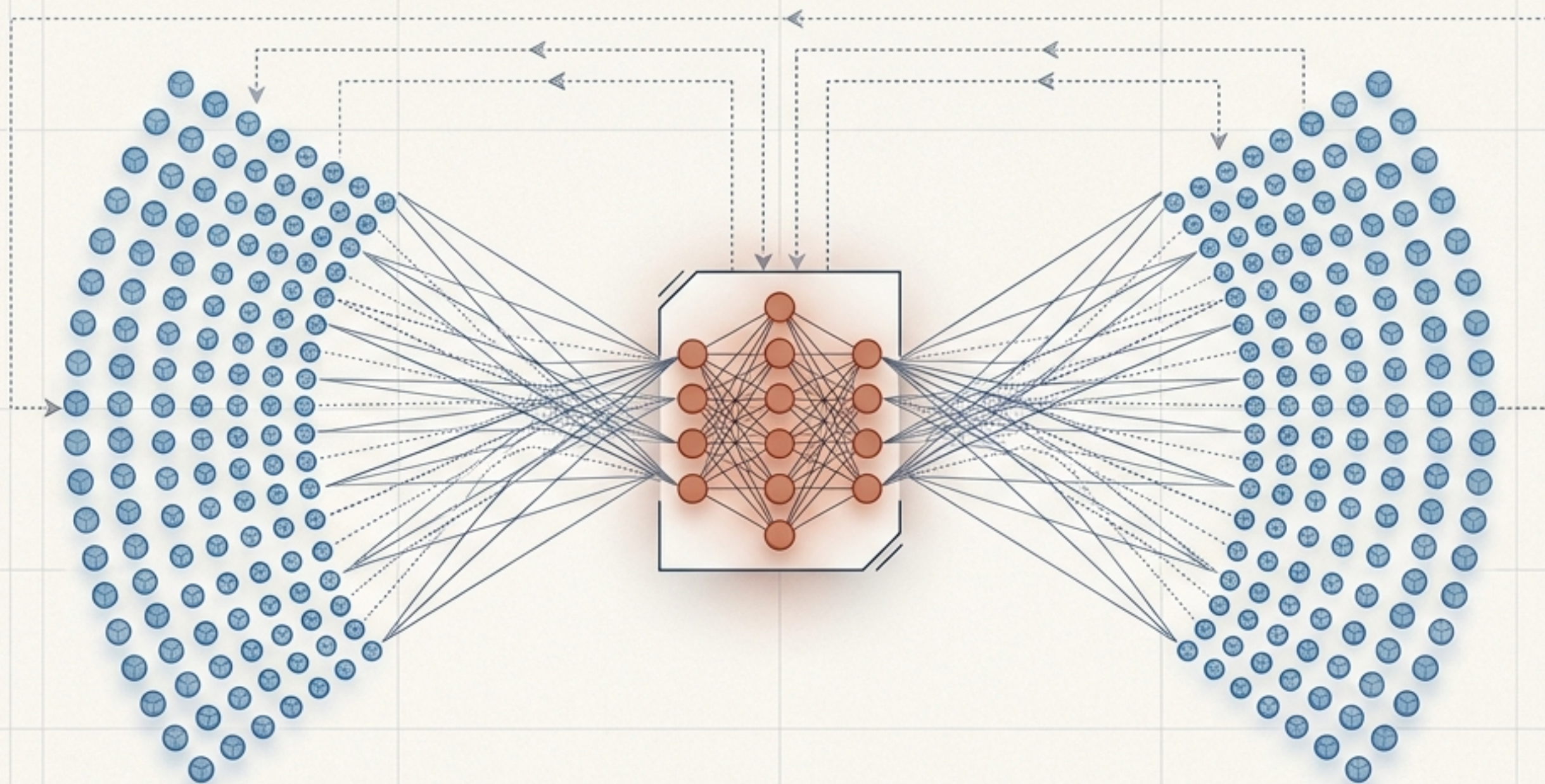




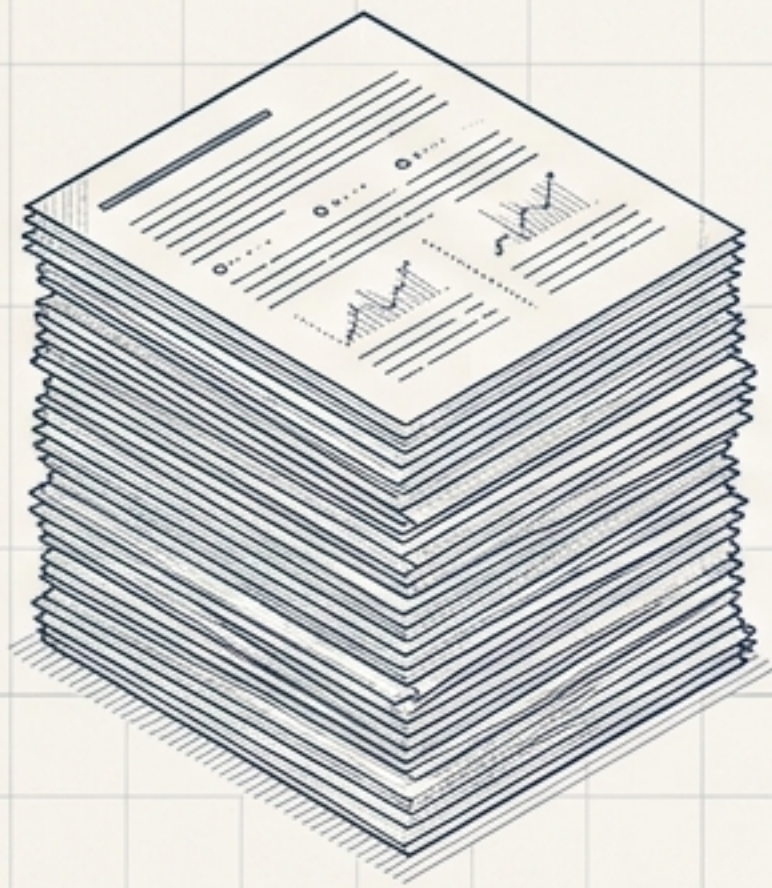
# Decoding Generative Autoencoders

From Mathematical Intuition to Neural Architecture

A Visual Synthesis



The Goal:  $\text{Input} \approx \text{Output}$



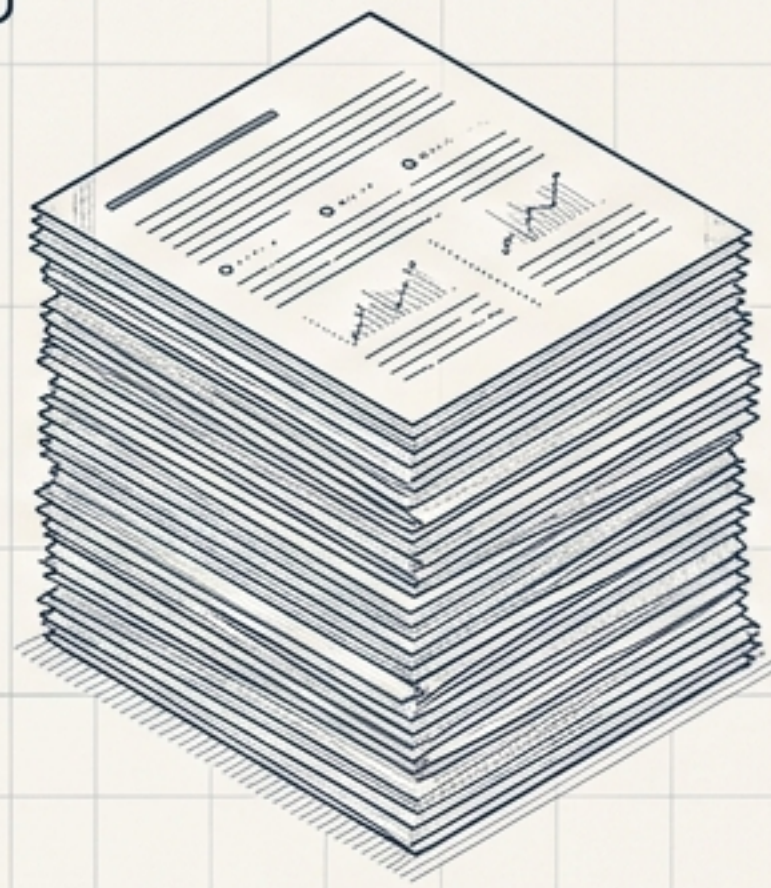
**Input**

(High-Dimensional Data:  
e.g. 784 pixels)



**The Zip File**

(Latent Representation)



**Output**

(Reconstructed Data)

### The Catch:

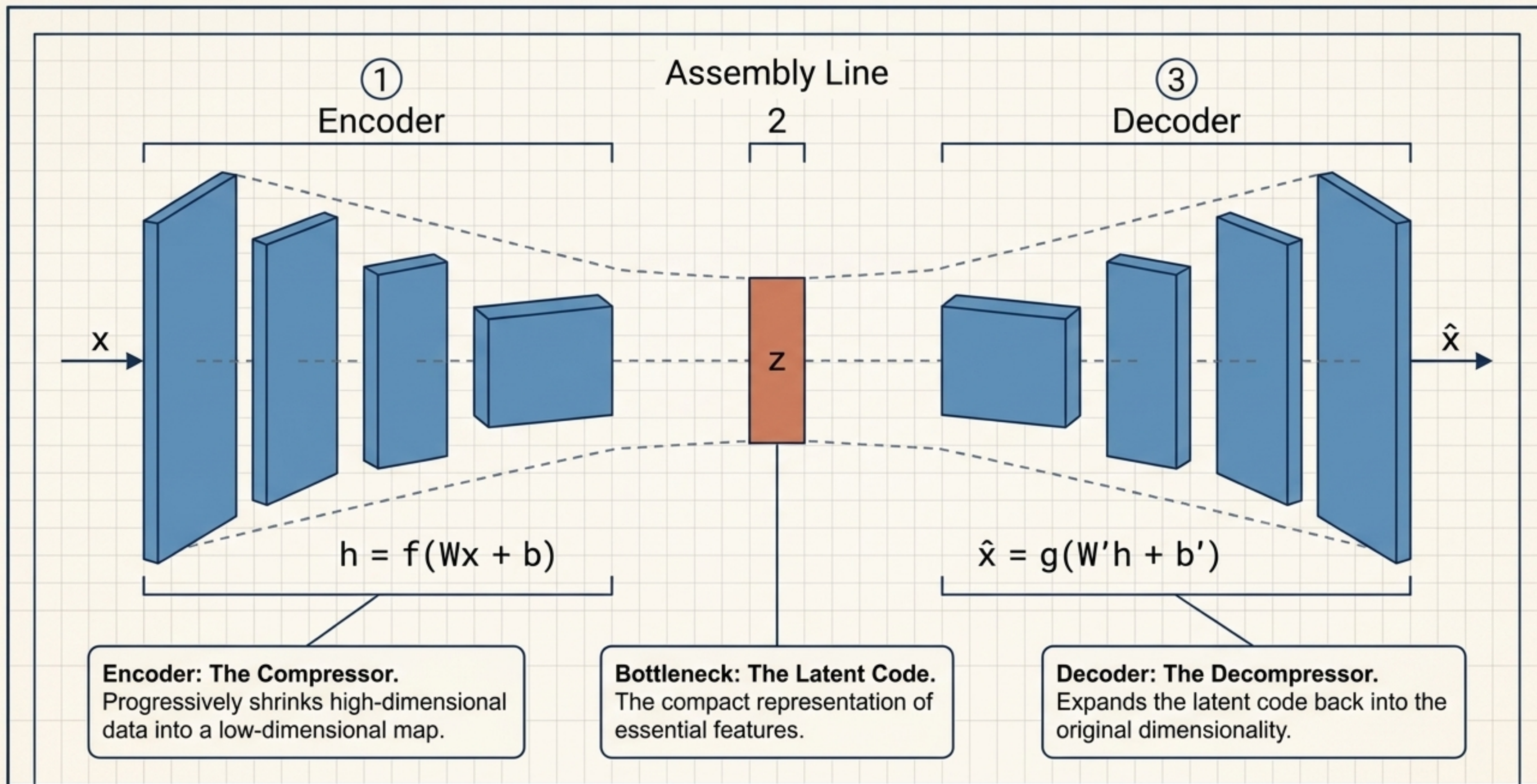
The network is not allowed to simply copy the data. It must learn the underlying patterns to rebuild it from scratch.

### The Key Insight:

The autoencoder asks, "What is the most important information needed to reconstruct this data?"

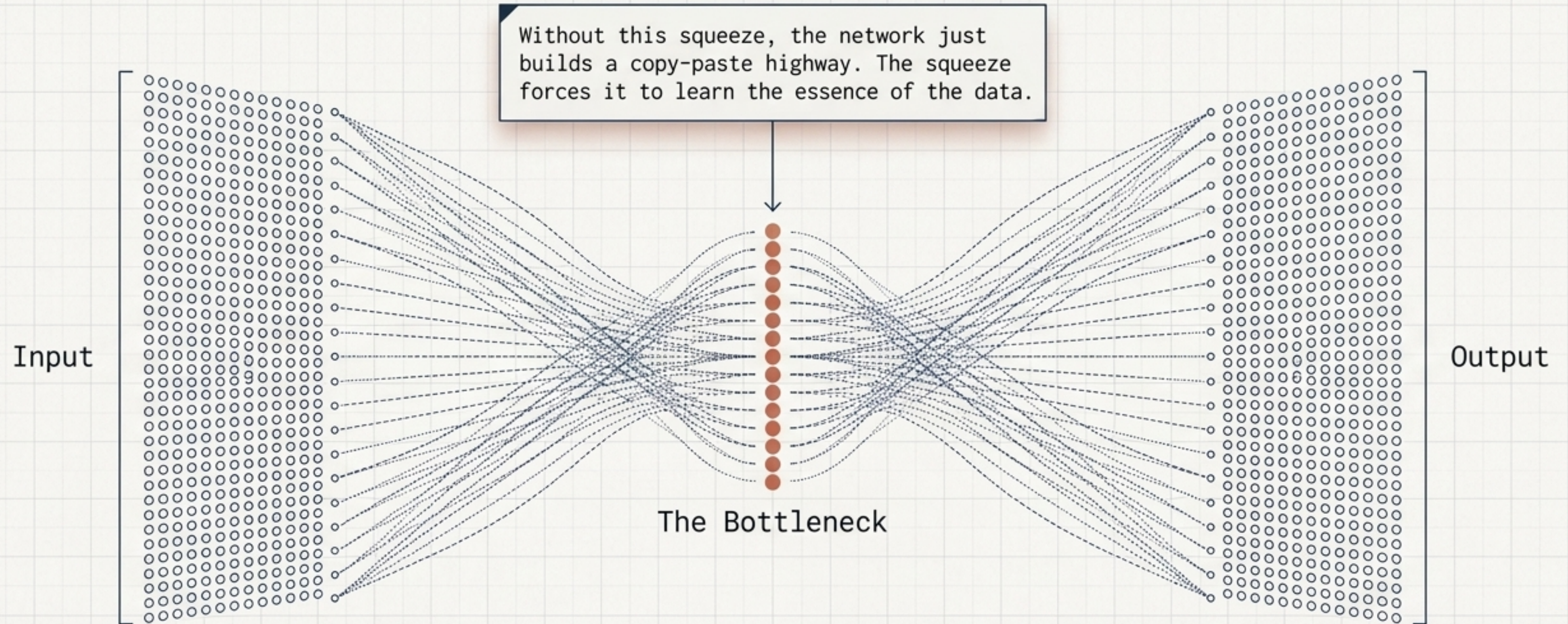


# AUTOENCODER ARCHITECTURE: A BLUEPRINT FOR DATA COMPRESSION





## THE NECESSITY OF THE BOTTLENECK: FORCING LEARNING THROUGH COMPRESSION

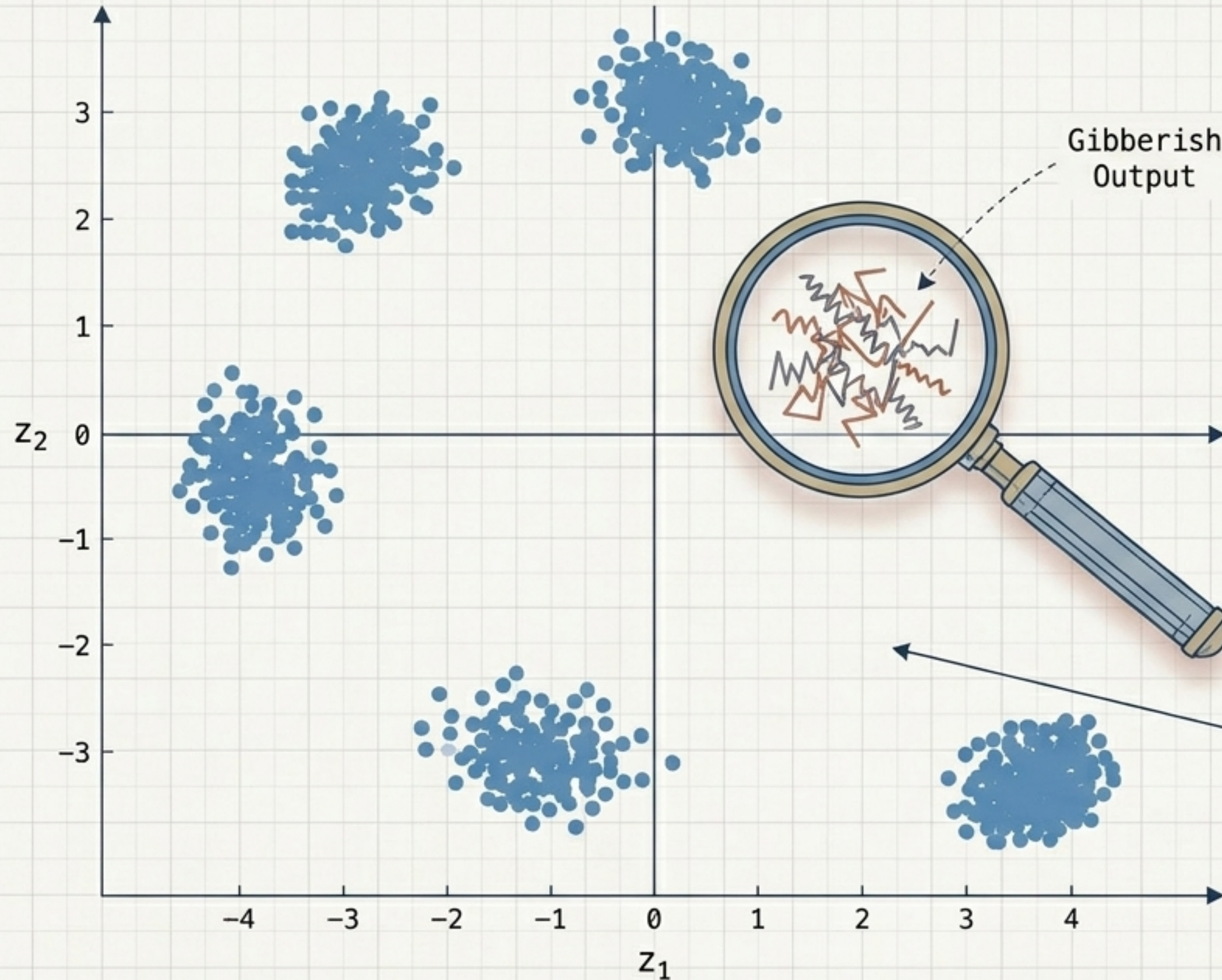


**Undercomplete Architecture:**  
Latent dimension < Input dimension.

**The Constraint:** By restricting the bandwidth, the network abandons noise and memorises only structural logic (e.g., 'a loop on top, a line on the bottom' for a number 9).



## THE LIMITATIONS OF DETERMINISTIC AUTOENCODERS: DISCONTINUOUS LATENT SPACE



### THE DETERMINISTIC WALL

Regular AEs assign one fixed  $z$  vector per input. Given the same input, you always get the exact same output.

### THE PROBLEM

The network only knows how to decode the specific dots it has memorised.

### THE DEAD ZONE

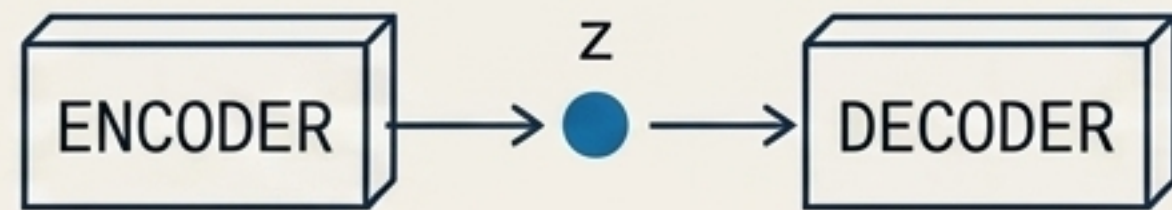
If you randomly sample a coordinate from the empty space and ask the decoder to build an image, it outputs pure gibberish.

You cannot generate new data.

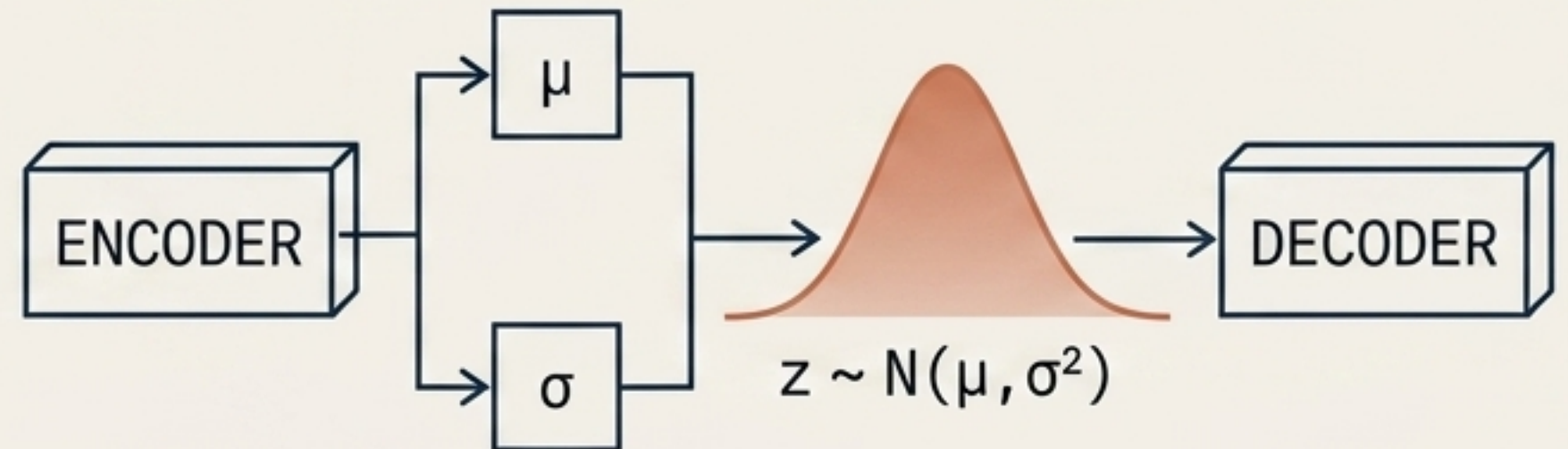


**The Probabilistic Paradigm Shift:** Instead of saying 'The code is exactly 3.5', the VAE says 'The code is likely around 3, give or take 0.5'.

### STANDARD AUTOENCODER



### VARIATIONAL AUTOENCODER (VAE)



$\mu$  (Mean): The centre of the distribution on the number line.

$\sigma$  (Sigma): The spread or width of the distribution.

The Result: By encoding data as overlapping clouds rather than isolated points, the latent space becomes continuous.



## THE ELBO TRADE-OFF: BALANCING FIDELITY AND STRUCTURE



**Reconstruction**  $E_q[\log p_\theta(x|z)]$ :  
How faithfully does the decoder  
rebuild the input  $x$ ?  
Ensures outputs look real.

**KL Regularisation**  $D_{KL}(q_\phi(z|x) || p(z))$ :  
How close is the learned distribution to  
a standard normal bell curve? Ensures the  
latent space stays smooth and structured.

Both terms must be minimised simultaneously to prevent collapse.



### The Anti-Collapse Guardrail.

Prevents the variance from shrinking to zero (which would turn the VAE back into a deterministic AE).

### The Constrictor.

Penalises the model if the distribution's spread gets wildly wide and chaotic.

$$D_{\text{KL}} = -\frac{1}{2} \sum \left( 1 + \log(\sigma_j^2) - \frac{\mu_j^2}{\sigma_j^2} - \sigma_j^2 \right)$$

### The Anchor.

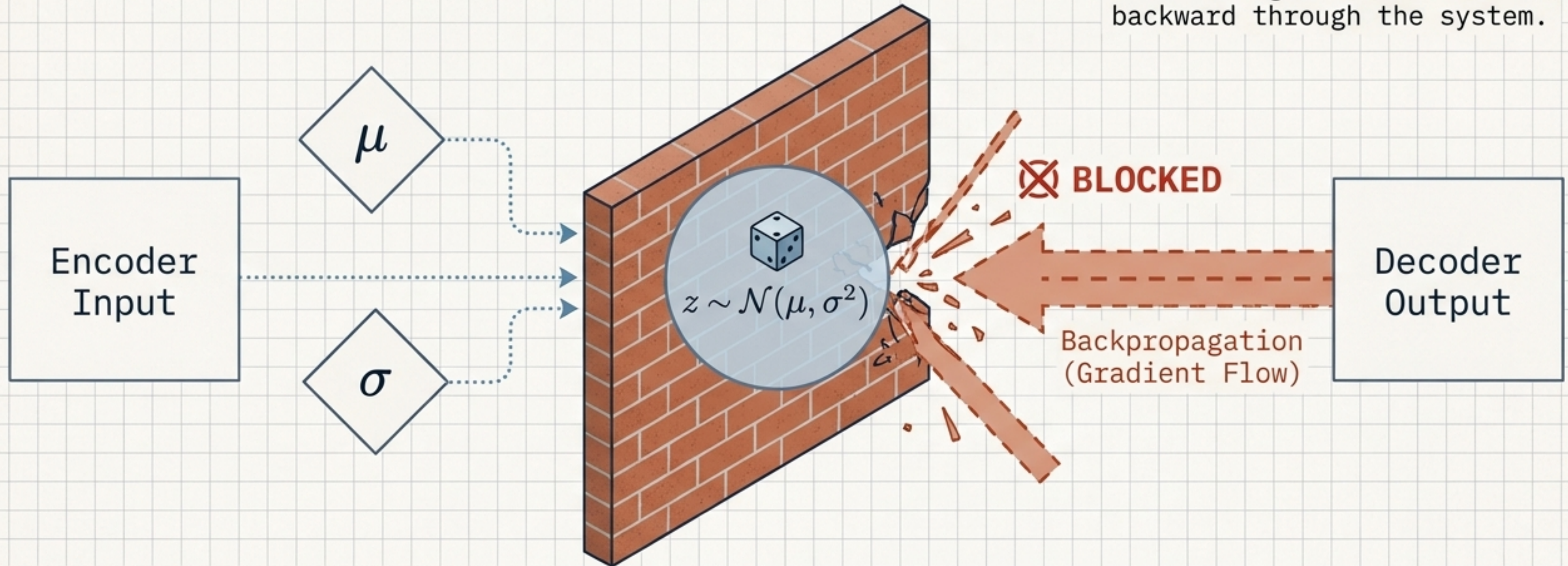
Penalises the model heavily if the centre of the distribution drifts away from 0.

**The Ultimate Goal:** Force the learned distribution  $q(z|x)$  to resemble the standard normal prior  $p(z) = \mathcal{N}(0,1)$ . If  $\mu=0$  and  $\sigma=1$ , the penalty is zero!



# THE STOCHASTIC ROADBLOCK: BACKPROPAGATION VS. RANDOM SAMPLING

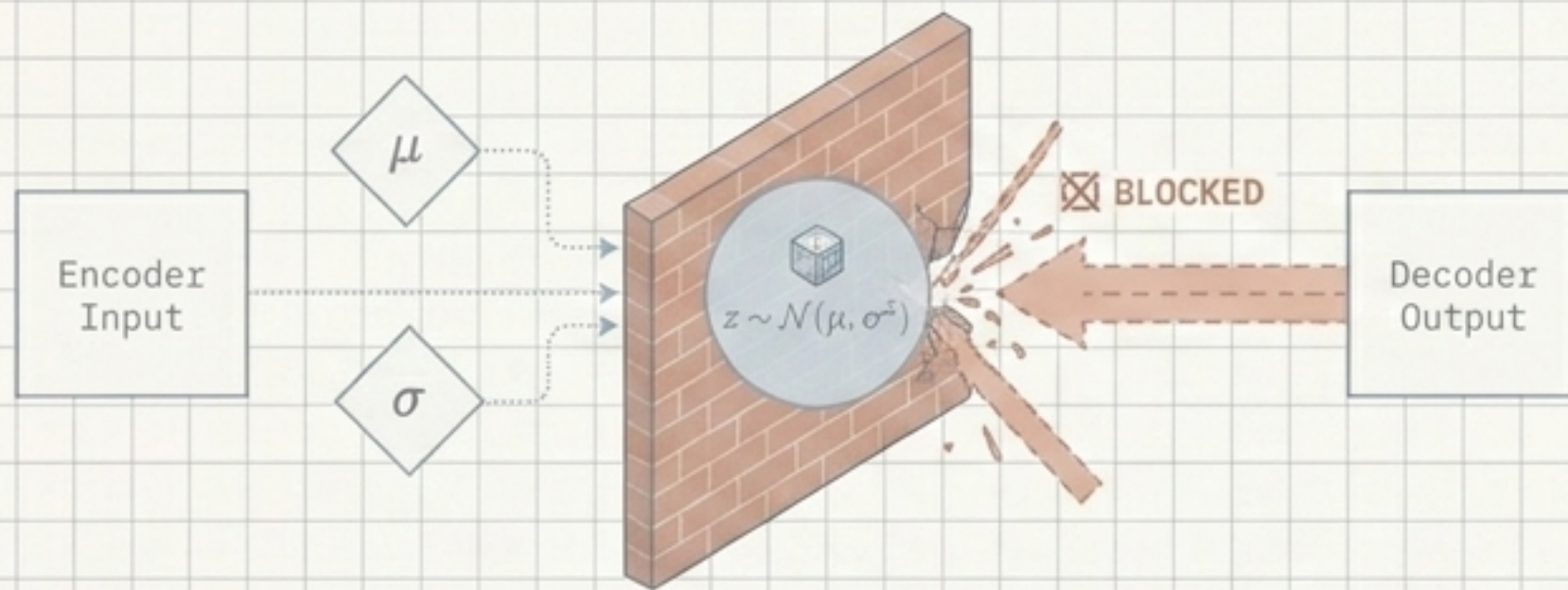
The Stochastic Roadblock:  
Neural networks learn by  
backpropagation—calculating  
the exact gradient of error  
backward through the system.



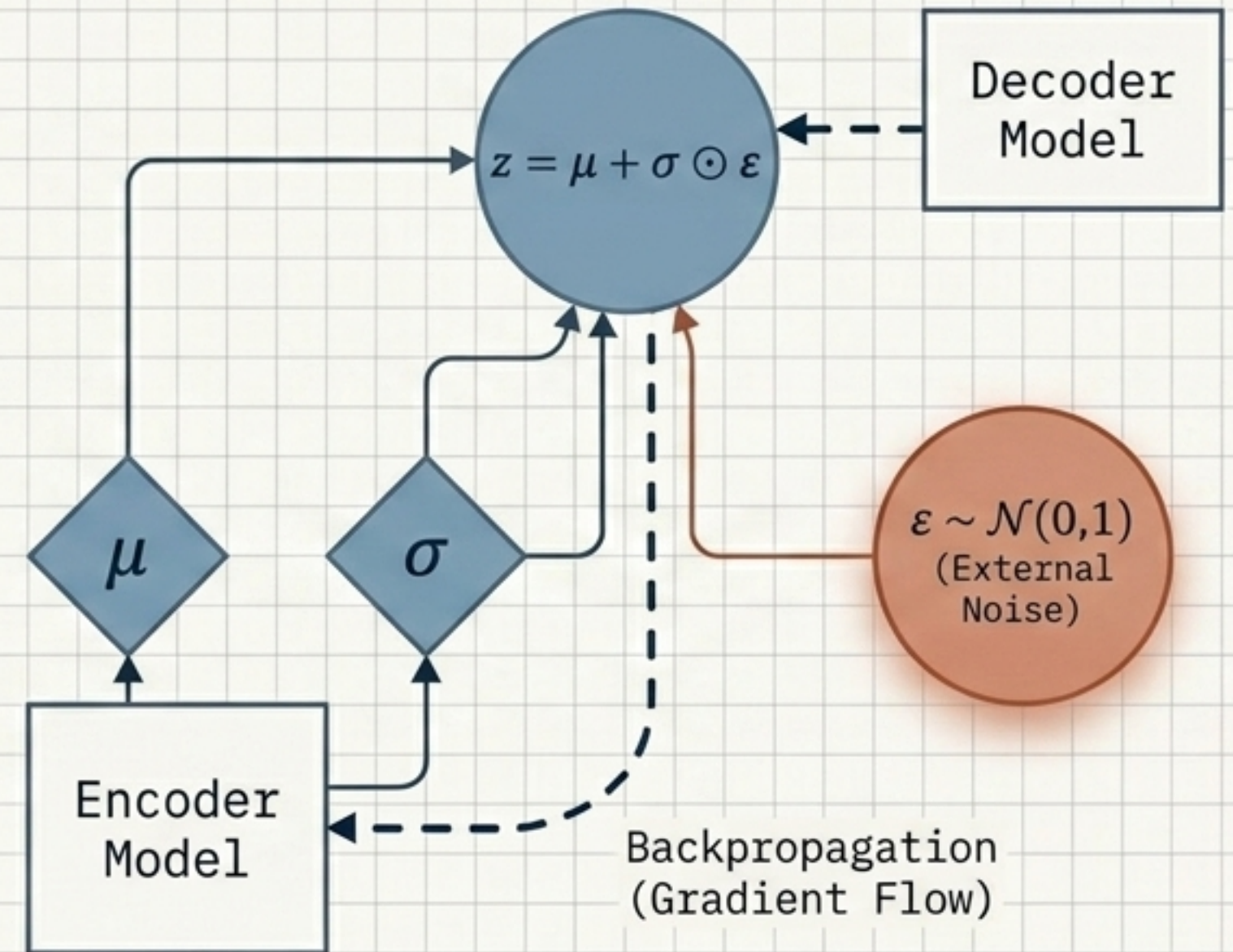
**The Conflict:** You cannot calculate the derivative of a random dice roll. Sampling  $z$  directly blocks the gradient flow. The network cannot learn.



## THE STOCHASTIC ROADBLOCK: BACKPROPAGATION VS. RANDOM SAMPLING



## THE BYPASS: THE REPARAMETERISATION CIRCUIT



**The Conflict:** You cannot calculate the derivative of a random dice roll. Sampling  $z$  directly blocks the gradient flow. The network cannot learn.

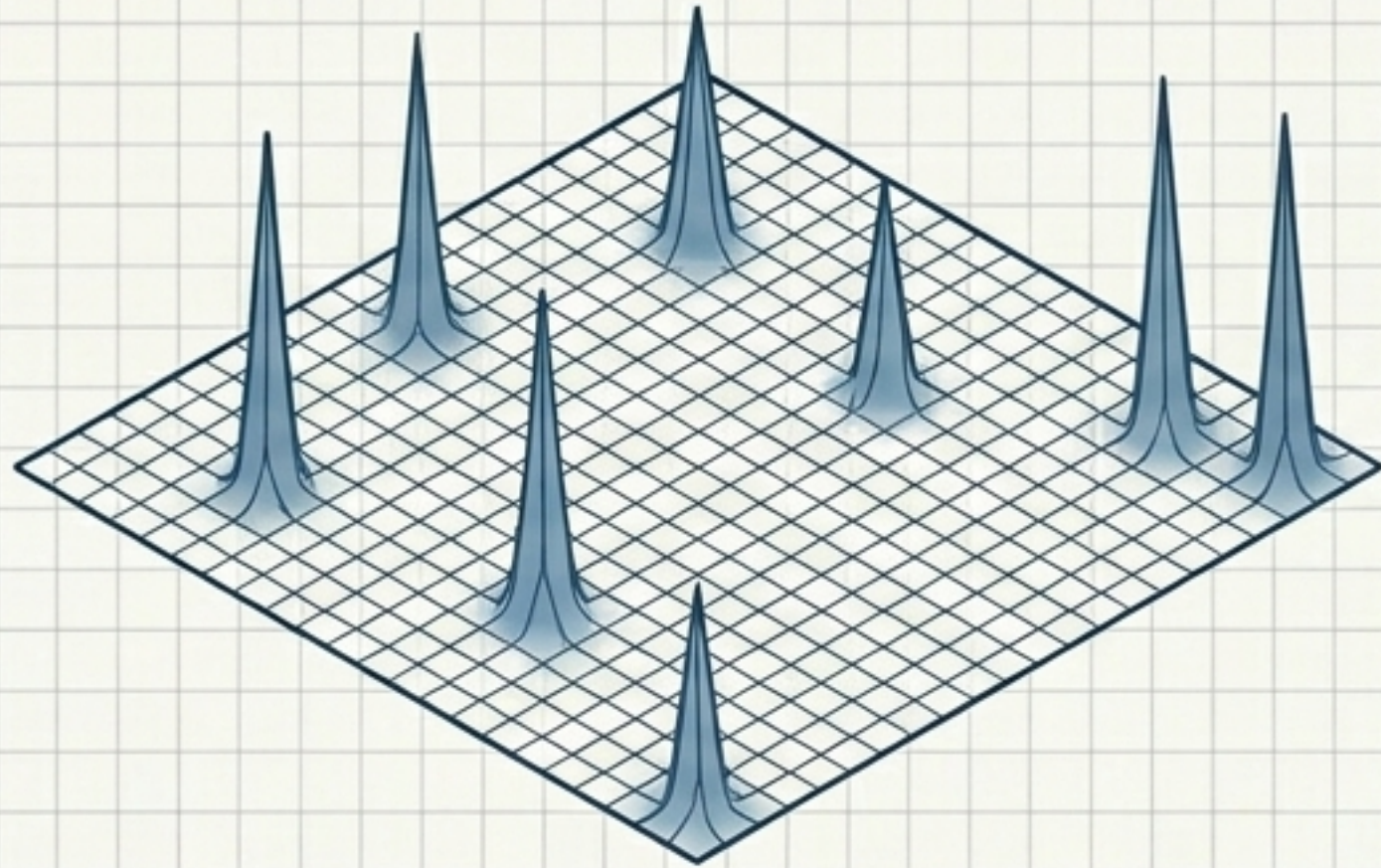
**The Reparameterisation Trick:** Instead of sampling  $z$  directly, we calculate it using an external source of randomness ( $\epsilon$ ).

**The Result:** The randomness is isolated.  $\mu$  and  $\sigma$  remain differentiable. Gradients flow, and the network learns!



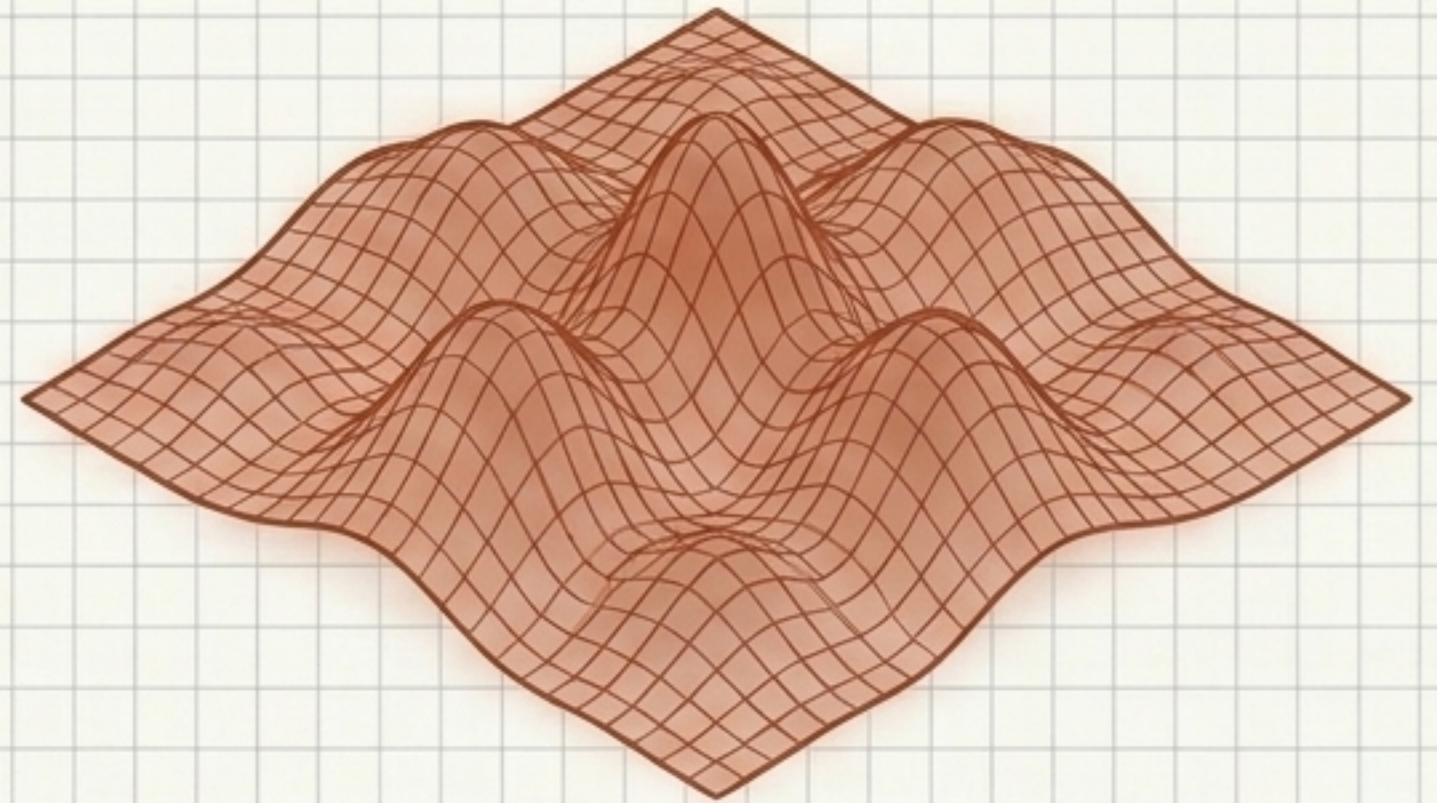
THE SYNTHESIS: GENERATION IS SIMPLY INTERPOLATION.

### UNREGULARISED AE



Gaps -> Gibberish

### REGULARISED VAE

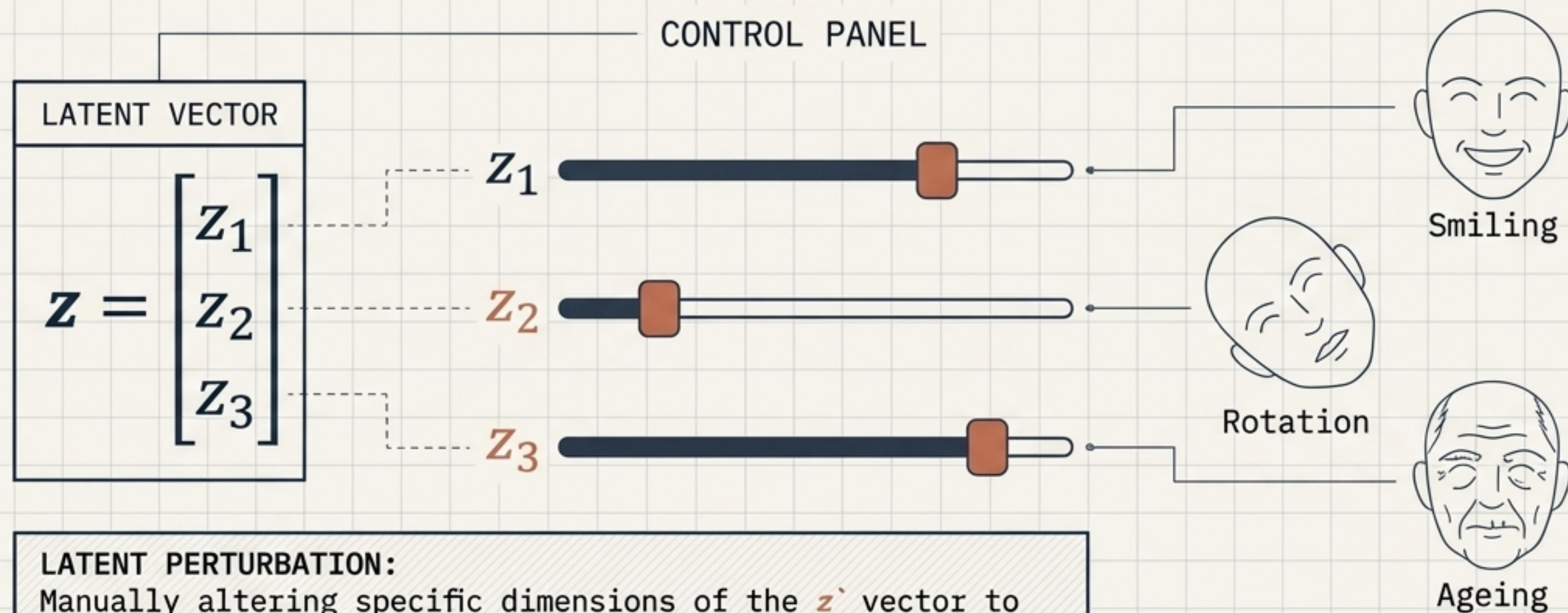


Overlapping -> Smooth Interpolation

**WHY IT WORKS:** By forcing the distributions to be soft and overlapping near the origin, KL divergence eliminates dead zones. You can now pick any random coordinate along the rolling surface and decode a realistic, novel output that blends the features of nearby data points.



# DISENTANGLEMENT IN VAEs: CONTROL VIA LATENT SPACE PERTURBATION



## LATENT PERTURBATION:

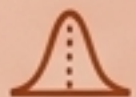
Manually altering specific dimensions of the  $\mathbf{z}$  vector to control specific features of the output.

## THE $\beta$ -VAE UPGRADE:

By increasing the weight ( $\beta$ ) of the KL Divergence term, the network is forced to 'disentangle' features. Each dimension of  $\mathbf{z}$  learns to control exactly one meaningful, independent trait without affecting the others.



# DIAGNOSTIC MATRIX: ARCHITECTURAL OVERVIEW

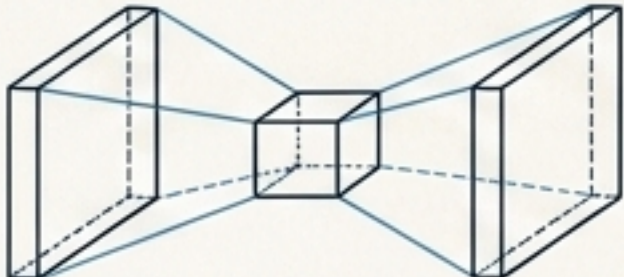
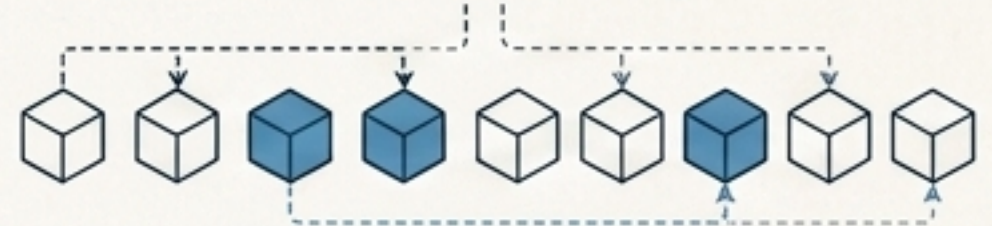
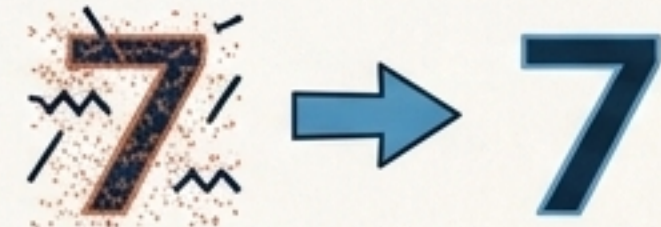
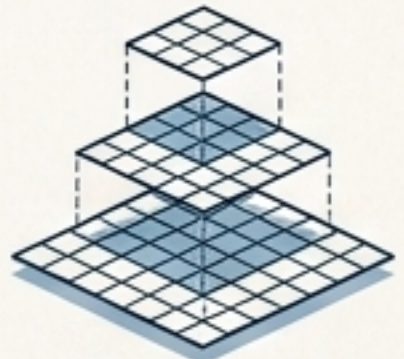
| ASPECT              | • Autoencoder •                                    |  Variational Autoencoder  | → → → Seq2Seq                                               |
|---------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Primary Goal        | Reconstruct Input                                  | Model Distribution & Generate                                                                                                                                                                   | Transform Sequence<br>$A \rightarrow B$                     |
| Input vs Output     | Same ( $\mathbf{x} \rightarrow \hat{\mathbf{x}}$ ) | Same ( $\mathbf{x} \rightarrow \hat{\mathbf{x}}$ )                                                                                                                                              | Different Domain<br>( $\mathbf{x} \rightarrow \mathbf{y}$ ) |
| Latent Space Nature | Fixed Point $\mathbf{z}$<br>(Deterministic)        | Distribution $\mathcal{N}(\mu, \sigma^2)$<br>(Stochastic)                                                                                                                                       | Context Vector $\mathbf{c}$<br>(Deterministic)              |
| Supervision         | Unsupervised<br>(Self)                             | Unsupervised<br>(Self)                                                                                                                                                                          | Supervised<br>(Parallel Corpus)                             |



**Key Diagnostic Check:** AEs and Seq2Seq both use encoder-decoder architectures, but AEs compress the same data domain, while Seq2Seq bridges different domains (e.g., English to French).

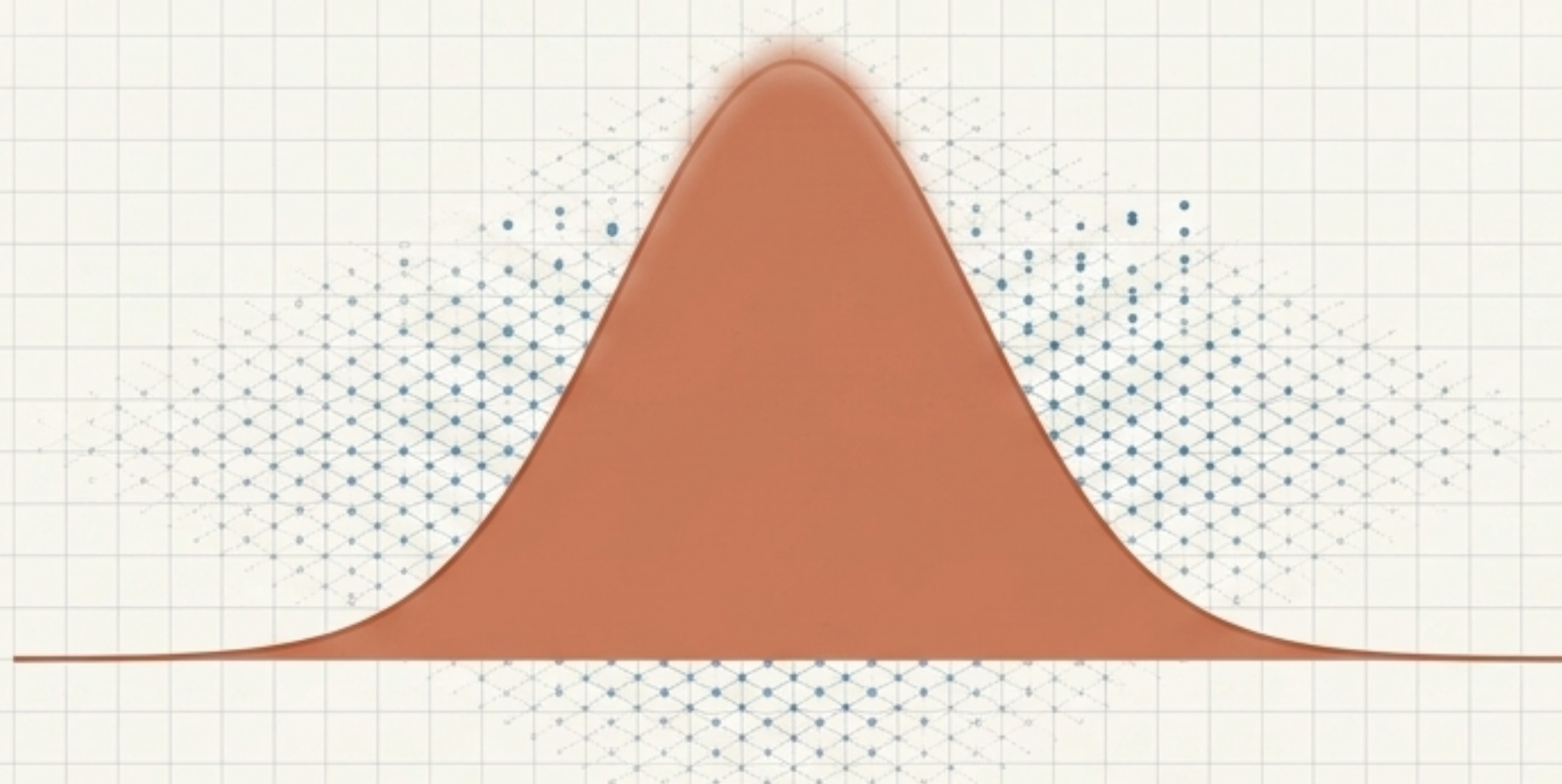


# THE AUTOENCODER ARSENAL: ARCHITECTURAL VARIANTS

| VARIANT             | CONSTRAINT                   | SCHEMATIC                                                                             | BEST USE                                        |
|---------------------|------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------|
| VANILLA             | Bottleneck                   |    | Basic dimensionality reduction.                 |
| SPARSE              | L1/KL Penalty on activations |    | Meaningful feature extraction.                  |
| DENOISING           | Input is corrupted           |  | Robust representation learning, removing noise. |
| CONVOLUTIONAL (CAE) | Spatial hierarchy            |  | Image compression and computer vision.          |

The architecture adapts to the constraint. What you force the network to struggle with determines what it learns.





# The Generative Engine

**Autoencoders do not exist merely to compress files.  
They exist to find the hidden mathematics of reality.**

By forcing data through a bottleneck, and smoothing that bottleneck with probability, VAEs learn the underlying blueprints of the world—allowing us to not just rebuild what exists, but to generate what could be.